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Exploring Human-AI trust in Industry 5.0: Insights from a Virtual Reality Intelligent Tutoring Case Study

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Abstract

Industry 5.0 promotes a human-centric approach to technological innovation, emphasizing collaboration between humans and intelligent systems. Within this paradigm, trust represents a fundamental condition for effective and reliable interaction. This study explores the dynamics of trust by integrating a theoretical analysis and an empirical case study. The first part of the research presents a literature-based examination of human–AI trust, identifying key influencing factors such as reliability, transparency, explainability, accountability, and ethical alignment. These insights informed the second phase of the study, which investigated a virtual reality (VR) training system equipped with a voice-based virtual assistant designed to support procedural skill learning. Participants completed a Trust in Automation (TiA) questionnaire covering six dimensions of trust, providing an overview of how users perceive reliability, clarity, and understanding within immersive training environments. Building on these findings, the paper discusses design considerations for developing trustworthy and human-centered VR training systems. It highlights the importance of calibrated feedback, communicative clarity, and transparency as essential elements for fostering trust and effective collaboration between humans and intelligent systems.

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1. Introduction

Industry 5.0 embodies a human-centric vision of digital transformation, emphasizing synergistic collaboration between humans and intelligent technologies to enhance productivity, creativity, and overall well-being. Within this paradigm, trust emerges as a crucial element that enables humans and artificial agents to cooperate effectively, share responsibilities, and act as complementary partners in complex environments. In recent years, immersive learning technologies have gained increasing attention as promising tools for industrial training. Virtual Reality (VR) environments, in particular, have proven effective in supporting skill acquisition and operational safety through simulation-based learning. More recently, Artificial Intelligence (AI) has begun to be integrated into such environments, giving rise to adaptive and interactive systems capable of providing personalized feedback, real-time guidance, and automated performance evaluation. Studies in this domain have highlighted the benefits of AI-driven VR systems in improving engagement, learning outcomes, and cognitive immersion [1,2]. Despite these advances, the role of trust between humans and AI in immersive training contexts remains critically underexplored. Existing research on human–AI trust has predominantly focused on autonomous or decision-support systems, where trust is conceptualized as confidence in algorithmic accuracy or decision reliability. However, far less is known about how trust develops when AI assumes an assistive or tutoring role, guiding rather than replacing human decision-making. Understanding how trust is established, maintained, and experienced in such guidance-oriented interactions is essential for advancing the human-centered vision of Industry 5.0. In these contexts, trust extends beyond technical performance. This study addresses this gap and provides valuable insights for the design of future AI-enhanced VR training environments that not only optimize skill development but also foster meaningful, trustworthy, and enduring human–machine relationships.

The remainder of this paper is structured as follows: Section 2 presents the theoretical background; Section 3 describes the materials and methods; Section 4 reports the results; Section 5 discusses the findings; Section 6 outlines key considerations; and Section 7 concludes with reflections and directions for future research.

2. Theoretical Background

Trust has long been recognized as a fundamental mechanism sustaining social relationships and cooperation. It has been examined across multiple disciplines, often described as a means of reducing uncertainty and enabling coordinated interaction among individuals. One of the most influential theoretical formulations conceptualizes trust as the willingness to be vulnerable to another party's actions, based on the expectation that the other will act in ways that are significant and beneficial to the trustor, even in the absence of direct control or monitoring [3]. This perspective highlights the relational and voluntary nature of trust, framing it as an intentional act of risk-taking grounded in positive expectations. Within this framework, perceptions of trustworthiness – including evaluations of competence, integrity, and benevolence – play a central role in shaping whether individuals engage in behaviors involving relational risk. Trust therefore emerges from both cognitive assessments of ability and moral alignment and affective beliefs concerning goodwill and intentions. Moreover, this conceptualization underscores the dynamic and iterative nature of trust, as prior experiences and feedback continuously influence the willingness to remain vulnerable over time. Although originally developed in organizational contexts, this theoretical perspective has become one of the most widely adopted foundations for understanding interpersonal trust more broadly. Its multidimensional structure, integrating cognitive and affective components, offers a robust basis for exploring how humans attribute reliability and moral value to others, and provides a useful lens for extending the notion of trust beyond human relationships – particularly in interactions with intelligent systems.

While trust between humans is grounded in perceptions of moral integrity and interpersonal intentions, its extension to interactions with AI requires a distinct conceptualization. In human–AI relationships, trust does not emerge from social reciprocity but from users' evaluations of the system's reliability, transparency, predictability, and competence. It reflects a primarily cognitive assessment of whether the AI behaves consistently, performs effectively, and communicates its operations in ways that users can understand. Empirical studies have explored these dynamics in various application domains. Some research has shown that trust in AI systems is shaped by multiple interrelated factors, such as the reputation or recognizability of the developer, which can implicitly convey reliability and accountability, as well as the perceived transparency and explainability of the system's operations. Users tend to trust

AI more when they can understand how it functions and why it produces specific outputs, whereas opacity or excessive autonomy can weaken confidence. Additional factors such as perceived usefulness, competence, and comfort during interaction further influence users' willingness to rely on AI [4].

Beyond these static determinants, several works have emphasized the dynamic nature of human–AI trust, showing that it evolves with users' experiences and system behavior over time. Research indicates that trust decreases following errors and only partially recovers once accuracy is restored, revealing an asymmetry between trust loss and recovery. The timing and severity of failures also matter – early or major errors tend to erode confidence more deeply than later or minor ones, as prior successful interactions may mitigate their impact. However, when systems acknowledge mistakes or provide transparent explanations, users are more inclined to rebuild confidence, underscoring the role of error communication and feedback in maintaining trust [5,6,7] .

Several studies have also investigated how the communication of uncertainty influences trust calibration. Findings consistently indicate that when AI systems clearly express their uncertainty, users develop more balanced and stable trust, avoiding both overreliance and complete distrust. Conversely, concealing or ambiguously communicating uncertainty leads to erosion of confidence as users struggle to interpret system reliability. The linguistic framing of uncertainty further shapes users' perceptions: first-person expressions (e.g., "I'm not sure, but…") evoke stronger emotional responses and encourage reflection, while more impersonal forms (e.g., "It's not clear, but…") are perceived as more neutral and professional. These findings collectively suggest that the style, clarity, and transparency of AI communication play an essential role in shaping trust [5,8,9].

Complementary evidence highlights the importance of user proficiency and understanding in building appropriate trust. Studies show that users with higher AI literacy are better able to interpret system behavior, discern competence, and calibrate trust accordingly, while those with lower proficiency may struggle to differentiate between reliable and unreliable outputs. Moreover, providing users with explicit information about the system's limitations can improve collaboration and understanding without reducing trust—suggesting that transparency about weaknesses fosters calibrated reliance rather than blind faith. Together, these insights emphasize that trust develops through continuous feedback and learning, where users refine their confidence based on observed performance and communication cues [10,11].

The communicative dimension of human–AI interaction has also been linked to trust development. Allowing users to engage in interactive explanations or conversational dialogues enhances comprehension and perceived control, which in turn strengthen trust. Clear, adaptive, and personalized communication helps bridge knowledge gaps and reinforces human confidence in AI-supported decision-making. These findings align with the broader recognition that trust emerges not only from system accuracy but also from how transparently and responsively the AI shares information throughout interaction [12].

Finally, the ethical and moral dimensions of AI behavior have been identified as crucial for long-term trust. Research demonstrates that when AI systems act in ways perceived as unethical or socially inappropriate, users' trust is significantly undermined – even if task performance remains unaffected. Moreover, conventional trust-repair strategies, such as apologies or denials, appear ineffective in restoring confidence after ethical violations, suggesting that ethical trust represents a distinct and particularly fragile dimension of human–AI relationships. Perceived integrity, fairness, and accountability therefore stand out as foundational elements of sustainable and meaningful trust in intelligent systems [13].

Building on these contributions, it becomes clear that trust in human–AI interaction is a multidimensional and evolving construct, shaped by a constellation of cognitive, communicative, affective, and ethical factors that operate in combination. On a cognitive level, trust depends on perceptions of accuracy, reliability, transparency, and comprehensibility, which enable users to interpret and anticipate system behavior. Communicative aspects—particularly clarity, explainability, and the expression and management of uncertainty—further influence how users evaluate the system's intentions and reliability, fostering more calibrated and sustainable forms of trust. From a social and affective standpoint, elements such as empathy, adaptability, and social presence contribute to a sense of collaboration and psychological safety, extending trust beyond purely functional appraisals. At the same time, ethical integrity, fairness, and accountability emerge as essential conditions for maintaining confidence over time, especially as AI systems take on roles that require alignment with human norms and values. Finally, trust in AI is inherently dynamic and context-dependent, evolving through experience, feedback, and the continuous calibration between overreliance and skepticism. Taken together, these insights emphasize that cultivating trustworthy human–AI

relationships require moving beyond technical performance toward a human-centered approach that integrates transparency, adaptive communication, emotional awareness, and ethical responsibility as core dimensions of design and interaction.

Although the literature provides a rich understanding of the mechanisms that influence trust in human–AI interaction, most of these insights stem from contexts where AI systems operate autonomously or as decision-support tools. As summarized in Table 1, trust in such settings is commonly conceptualized through multiple interrelated dimensions which encompass factors such as accuracy, transparency, explainability, empathy, and accountability. In these scenarios, trust is often framed as confidence in algorithmic judgment or performance reliability. However, when intelligent systems are embedded within training or guidance-oriented environments, the nature and meaning of trust may shift.

Table 1. Key dimensions shaping users' trust in human–AI interaction as emerged from the reviewed studies.

Dimension	Main Factors	Key Insights	Ref.
Cognitive / Functional	Accuracy, Reliability, Transparency, Explainability, Comprehensibility	Users trust AI more when they can understand how and why it operates reliably and consistently. Trust increases with clarity of system logic and perceived technical competence.	[4], [5], [6], [7]
Communicative	Clarity, Uncertainty Communication, Feedback, Conversational Explainability	Clear communication, explicit expression of uncertainty, and opportunities for dialogue foster calibrated and sustainable trust.	[5], [8], [9], [12]
Affective / Social	Empathy, Adaptability, Social Presence	Emotional attunement, adaptive behavior, and perceived social presence enhance comfort and psychological safety in interaction.	[4], [12]
Ethical / Accountability	Ethical Integrity, Fairness, Responsibility, Trust Repair	Perceptions of moral integrity, fairness, and accountability sustain long-term confidence; ethical violations severely damage trust.	[13]
Dynamic / Contextual	Feedback, Error Handling, Trust Calibration, Experience Over Time	Trust evolves through interaction experience, feedback, and error management; calibrated trust prevents overreliance or distrust.	[6], [7], [10], [11]

In skill development and assistance contexts – such as VR-based industrial training – AI is not expected to replace human judgment but to support learning, scaffold performance, and provide feedback. Here, trust becomes less about delegating decisions and more about accepting guidance, perceiving competence, and maintaining engagement throughout the interaction. This raises a critical question: *"If AI in such environments must be explainable, transparent, ethical, and accountable, what do these principles concretely mean for the design of systems whose purpose is to teach, assist, or guide rather than decide?"*

Addressing this question requires translating the abstract dimensions of trust identified in the literature into practical design and interaction strategies that can sustain credible, empathetic, and ethically aligned relationships between human learners and intelligent virtual assistants within immersive training contexts.

3. Material and Methods

Building on the theoretical foundations of trust in human–AI interaction, this study explores how these principles can be operationalized within an immersive VR-based industrial training environment. The system was designed to integrate a virtual assistant capable of providing real-time, context-aware guidance through natural language

interaction. The following subsections detail the training task, technical infrastructure, and interaction pipeline developed to investigate how users experience and calibrate trust in such settings.

3.1 Training task definition

The interactive virtual system aims to reproduce a realistic industrial environment (a maintenance and alarm management task in a foundry) in which the operator must both follow a predefined operational workflow and make decisions based on procedural logic, while a virtual assistant provides a procedural guidance through vocal request–response query interaction. The operational workflow is composed of several sequential steps, each associated with decision points and context-dependent actions. The purpose of a virtual assistant is to deliver real-time, context-aware guidance throughout the operational procedure, effectively tutoring operators as they complete their assigned tasks within an immersive training environment. When operators experience uncertainty or wish to confirm their next action, they can request assistance directly through the VR interface using natural language queries. Moreover, establishing user trust in the assistant's responses is essential for the system's effectiveness. Therefore, the case of study is oriented to how operators perceive trust in the interactive environment based on key dimensions such as reliability, accuracy, transparency, clarity of communication, and the quality of feedback provided by the AI assistant.

3.2. VR Interface platform & LLM preparing

UPBGE, an open-source fork of the Blender Game Engine that supports embedding game logic and Python scripting directly into Blender scenes, was selected as the development platform for the VR environment. Its open-source architecture and high degree of integration flexibility with external tools make it well suited for experimental prototyping. The embedded Python interpreter enables seamless scripting and access to the full Python library ecosystem, facilitating communication with external applications. Moreover, UPBGE inherits Blender's core functionalities, including 3D modeling, animation, and physics simulation, as well as full OpenXR support for Oculus Meta Quest devices. These features are essential for accurately replicating industrial environments and delivering high-fidelity, interactive training scenarios in VR.

To ensure low-latency request–response interaction with the language model, LM Studio was selected as the local model hosting environment. Operating as a local server, LM Studio minimizes latency and memory overhead while supporting natural, conversational behavior suitable for interactive guidance. This configuration also enables precise control over the system prompt, contextual variables, linguistic style, and response constraints, ensuring consistent and domain-appropriate feedback. Additionally, LM Studio provides integrated model management, offering access to a wide range of open-source large language models (LLMs) hosted on Hugging Face, a major platform for machine learning and natural language processing resources.

For this prototype, the chosen LLM was OpenHermes 2.5 – Mistral 7B, a model optimized for instruction-following and conversational assistance. This model was selected for its strong performance in training and coaching contexts, multilingual support, and efficient inference speed on mid-range GPU hardware. Balancing response quality, computational efficiency, and local autonomy, OpenHermes 2.5 – Mistral 7B is particularly well suited for real-time AI tutoring within VR-based industrial training environments.

3.3 Situational context and system prompt

To ensure coherence and contextual accuracy in the model's responses, LM Studio enables the configuration of a situational context that supports domain-specific and consistent guidance during training interactions. This is achieved through a system prompt that defines the operational framework, boundaries, and objectives guiding the language model's output. The prompt specifies key parameters such as role definition, linguistic tone, technical vocabulary, and the desired level of formality. Once established, it remains active throughout the interaction, thereby ensuring consistency and alignment with the procedural and instructional requirements of the VR-based industrial environment. In addition, a structured JSON file was developed to define the task context, including the number of steps, task name, and detailed descriptions of each procedure. This file provides the contextual information necessary for guiding the operator's training process. By organizing procedural data in this format, both the VR application and the language model can access and interpret a shared task framework, enabling accurate and consistent step-by-step assistance during the training session.

3.4 Human-AI vocal interaction engineering in the VR training system

An overview of the VR training system is presented in Figure 2. Upon entering the VR environment through the head-mounted display (HMD), the user is represented by a first-person avatar with fully animated hands and fingers. Interaction with the intelligent assistant is initiated by pressing a button on the heads-up display (HUD), which activates the audio capture module and records the user's verbal input via the integrated microphone. The recorded speech is processed by a Python-based middleware that transmits the audio stream to the Google Speech Recognition API for transcription. The resulting text serves as input to the OpenHermes 2.5 – Mistral 7B language model, hosted locally in LM Studio. Guided by the system prompt, the model interprets the user's intent and generates a contextually appropriate textual response. This output is then converted into natural-sounding speech using the Google Text-to-Speech (gTTS) API and reintegrated into the VR environment as an audible response. Through this bidirectional pipeline, the system supports seamless, voice-based interaction with the virtual assistant, enabling adaptive, context-aware guidance and an immersive training experience.

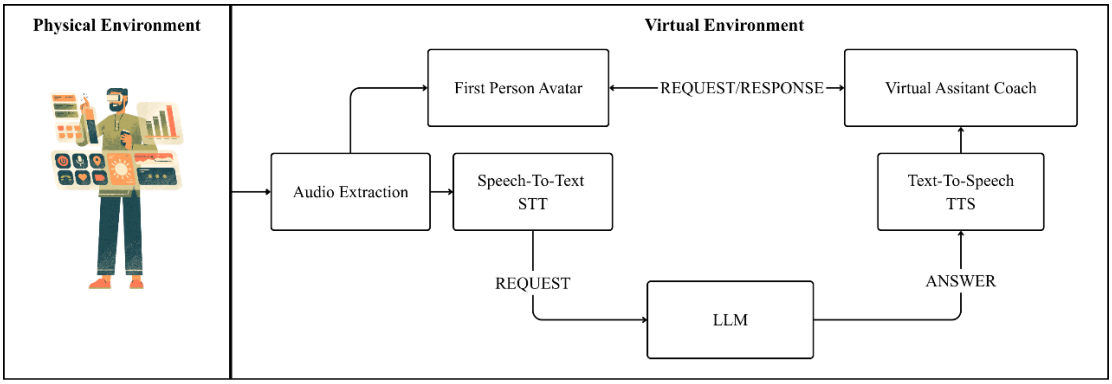


Figure 2. VR-assisted training system architecture

3.4.1 VR Interface Platform & LLM Integration

A Python script embedded within UPBGE was developed to establish a communication bridge between the VR environment and LM Studio. This connection was implemented through a REST API (Representational State Transfer Application), compatible with the OpenAI chat-completion endpoint, enabling real-time exchanges between the VR interface and the locally hosted language model. Communication occurs via the HTTP POST protocol, through which contextual task data and user queries are transmitted to the LLM, and corresponding responses or instructions are returned. This integration supports dynamic, conversational guidance, allowing the AI assistant to deliver step-by-step support to operators during industrial procedure simulations.

3.4.2 Hardware Requirements

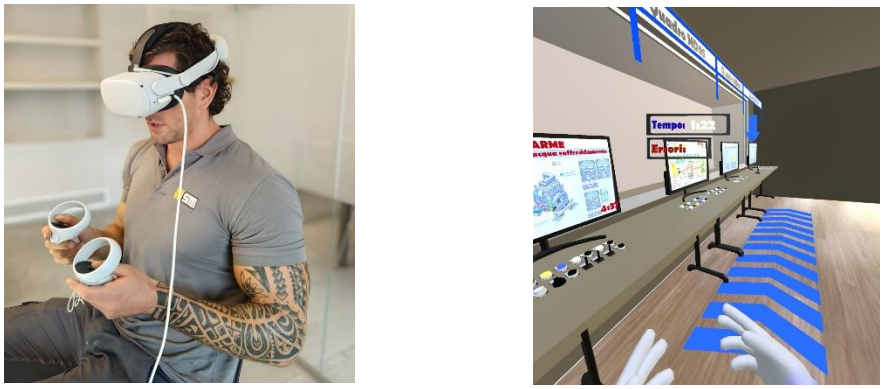
The system was deployed on a high-performance computing platform to ensure seamless VR integration, real-time AI-driven interaction, and immersive user experiences. The hardware configuration consisted of a Lenovo Legion Pro 5 laptop equipped with an Intel Core i7-14700HX processor (16 cores, 24 threads) and an NVIDIA GeForce RTX 4060 GPU (8 GB GDDR6 VRAM, PCIe 4.0). The system also included 32 GB of DDR5 RAM, supporting efficient data processing and high-quality graphical rendering. Virtual reality deployment was implemented using the Oculus Meta Quest 2 headset.

3.5 Case study and data collection with TiA questionnaire

The developed system was evaluated with a group of four industrial operators from a metallurgical company, selected to represent a range of ages, nationality, work experience, and role. Although the participants spoke different native languages, all were able to interact with the AI-assisted training system, which processed and interpreted their verbal inputs effectively. Each participant possessed prior hands-on experience with the industrial workflow procedures simulated in the VR environment, ensuring that the evaluation reflected realistic operational conditions.

During the experimental phase, the interactive VR system automatically recorded the completion times of the training sessions and procedural error rates to quantitatively evaluate user performance as shown in Figure 2. These metrics provided insight into both the effectiveness of the AI-assisted training environment and the level of trust and adaptability demonstrated by operators when interacting with the system. All participants were asked to review and sign an informed consent form, acknowledging their voluntary participation and understanding of the study's objectives and procedures. Each participant was instructed to perform the same operational workflow they normally execute in the real system, but this time within the virtual training environment. At the beginning of the session, the operators were guided through an introductory tutorial that explained the use of voice commands and system interaction protocols. This ensured that all participants understood how to request assistance and navigate the training workflow before starting the evaluation.

After completing the VR-based training session, all participants were asked to complete the Trust in Automation (TiA) questionnaire, adapted into a Google Forms format for ease of distribution. The TiA instrument, developed by Körber [14], consists of 19 items designed to assess different dimensions of human trust toward automated systems, including reliability, predictability, faith, and perceived safety. As a way of measurement, the TiA uses a Likert-scale, which is based on summative scaling, where respondents use a ranked scale to indicate their agreement with statements. The items to evaluate are organized by scales: Reliability/Competence, Understanding/Predictability, Familiarity, Intention of Developers, Propensity of Trust and Trust in Automation.



(a) (b)

Figure 2. Operator wearing a virtual reality headset during the simulation (a) and corresponding internal view of the headset (b).

4. Results

The Trust in Automation questionnaire provided meaningful insights into how participants perceived and developed trust within the VR training environment. It assessed six dimensions each representing a fundamental aspect of human–technology interaction. As illustrated in Figure 3, the lowest average score was recorded for *Familiarity*, whereas *Trust in Automation* obtained the highest.

The comparatively low rating for *Familiarity* likely reflects the novelty of the VR environment and participants limited prior exposure to similar training systems. This lack of experience may have constrained users' initial confidence and comfort, as familiarity is often associated with a sense of control and predictability in interaction. Conversely, the high score for *Trust in Automation* suggests that participants perceived the system as dependable and supportive throughout the training experience. Consistent system performance, clear instructions, and responsive feedback likely acted as trust-enabling cues, compensating for the absence of prior experience and fostering a generally positive trust orientation. These findings indicate that even without an established experiential background, users can still form a robust sense of trust when the system demonstrates reliability, clarity, and effective guidance.

When interpreted through the lens of the theoretical framework presented in the background section, several meaningful correspondences emerge. The literature on human–AI trust emphasizes that confidence in technological systems stems from a combination of technical, communicative, and ethical dimensions – specifically, competence, transparency, explainability, and accountability. The empirical results of the TIA questionnaire mirror these

conceptual pillars in practice. High evaluations for *Reliability/Competence* and *Trust in Automation* align with the theoretical constructs of competence and predictability, reflecting users' appreciation of stable and accurate system behavior. The moderate ratings for *Intention of Developers* and *Understanding/Predictability* echo users' awareness of transparency and explainability, suggesting that they value systems whose functioning and intentions are intelligible and coherent. Meanwhile, the lower *Familiarity* scores reinforce the importance of experience and feedback—identified in the literature as prerequisites for developing calibrated trust over time.

Taken together, these insights demonstrate how empirical evidence substantiates and extends theoretical perspectives on trust in human–technology interaction. While previous research has primarily focused on explainability and transparency as key enablers of human–AI trust, the present findings show that similar mechanisms operate even in non–AI-based contexts such as VR training. Users' trust arises not solely from the system's technical performance but also from its clarity, consistency, and perceived guidance—characteristics that embody the same human-centered values underlying trustworthy AI. This convergence between theoretical and practical perspectives highlights that trust, whether directed toward a human, an AI, or a virtual system, is a dynamic construct shaped by continuous interaction, interpretability, and experiential learning.

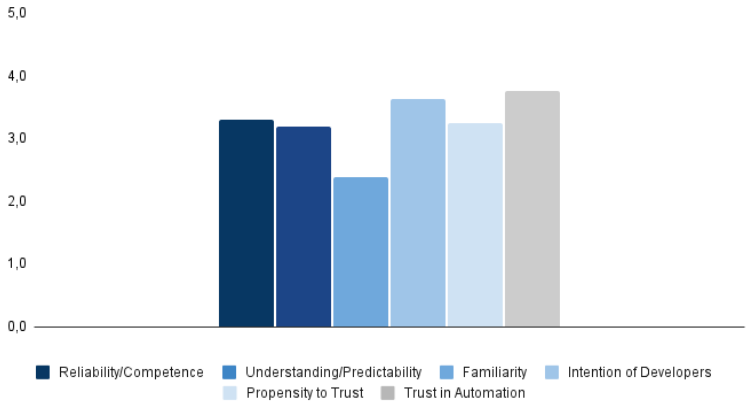


Figure 3. Average scores assigned to the TiA questionnaire factors.

5. Discussions

The insights derived from both the theoretical framework and the case study allow for a deeper reflection on how the key factors influencing human–AI trust – summarized in Table 1 and empirically observed through the TiA questionnaire results – can be concretely translated into design principles for immersive VR training environments. In such contexts, the virtual assistant does not act as a decision-maker but rather as a tutor and facilitator, supporting learners through demonstration, feedback, and adaptive guidance. Consequently, trust in this setting shifts from confidence in autonomous performance to confidence in the credibility, clarity, and fairness of the learning process itself. From a cognitive perspective, reliability and transparency constitute the foundation of trustworthy interaction. The system should provide feedback that is consistent, traceable, and meaningfully connected to user actions. For example, when an operator performs a procedural step incorrectly, the assistant could identify the specific mistake, explain its cause, and link it to relevant safety standards or operational goals. Such clear and contextualized explanations strengthen perceptions of competence and accountability, transforming the system from an opaque evaluator into a reliable learning partner.

On a communicative level, the assistant's language, timing, and level of detail are crucial to maintaining calibrated trust. Explanations should be context-sensitive—delivered at the appropriate moment and tailored to the learner's expertise. Novice users may require step-by-step guidance, while experienced operators may prefer concise feedback or performance summaries. The assistant can also convey calibrated uncertainty by offering alternative suggestions (e.g., "You might also try adjusting the angle slightly") when multiple solutions exist. Communicating uncertainty in this way prevents overreliance and promotes reflective engagement, aligning with the principle of trust calibration identified in the theoretical framework.

The affective and social dimensions of trust become particularly salient in VR due to the immersive and embodied nature of the experience. Nonverbal cues – such as tone of voice, gaze direction, or posture – can convey empathy and attentiveness, making users feel supported rather than evaluated. For instance, an assistant that modulates its behavior in response to user performance (e.g., offering encouragement after repeated errors) can reinforce emotional trust and sustain motivation, fostering a sense of psychological safety within the virtual learning environment.

From an ethical and accountability standpoint, fairness and transparency in evaluation are essential. The system should clearly communicate assessment criteria and ensure consistent treatment across users. For example, two learners who make the same mistake should receive equivalent feedback, regardless of their prior performance or background. Providing traceability of assessment logic and access to performance data enhances users' perceptions of fairness, responsibility, and system integrity – factors directly tied to ethical alignment within the trust framework.

Finally, trust in VR-based training systems is inherently dynamic and should evolve with user competence. Adaptive mechanisms that adjust feedback over time can help sustain trust while encouraging autonomy. In early stages, frequent corrective guidance may be necessary; later, the system can prompt self-reflection (e.g., "Do you think this step met the safety criteria?"), promoting a sense of partnership and shared learning. Such progressive adaptation supports calibrated trust – a balanced confidence grounded in both system reliability and user growth.

Collectively, these reflections underline several practical implications for the design of trustworthy, human-centered VR training systems. First, trust should be treated as an interactive and evolving process, nurtured through feedback, transparency, and adaptive communication. Second, explainability must serve a pedagogical function – helping users understand not only how the system provides guidance but also why. Third, emotional and ethical dimensions are not secondary but integral to trust: a credible virtual tutor should be competent, transparent, empathetic, and fair. Ultimately, translating the abstract principles of transparency, reliability, and ethical responsibility into tangible interaction features embodies the human-centric vision of Industry 5.0, where intelligent technologies act as collaborative partners that enhance learning, autonomy, and well-being.

6. Conclusions

This study combined a theoretical exploration of human–AI trust with an empirical investigation conducted through a VR-based training case study. By integrating insights from the literature with the results of the Trust in Automation questionnaire, the work identified and contextualized the main factors shaping trust in human–technology interaction—such as reliability, transparency, explainability, and experience—and discussed how these can be operationalized in the design of immersive training environments.

The findings demonstrated that even in the absence of explicit AI components, users can develop a meaningful sense of trust toward virtual systems when clarity, consistency, and fairness are ensured. The reflections proposed on cognitive, communicative, emotional, and ethical dimensions offer a foundation for designing future AI-driven VR tutors that embody human-centered trust principles aligned with the vision of Industry 5.0.

However, the study presents some limitations, particularly related to the limited number of participants, which constrains the generalizability of the findings. Future work should expand the sample size and involve multiple companies to include users with diverse backgrounds, native languages, and levels of experience – both with VR technologies and with the specific industrial procedures simulated.

Furthermore, a promising direction for future research involves the integration of artificial intelligence into the VR training system, enabling adaptive and personalized tutoring. Applying the design factors identified in this study – transparency, reliability, accountability, and ethical alignment – will be crucial to ensure trustworthy collaboration between humans and intelligent systems in next-generation industrial learning environments.

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